YAN CHEN

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WORK EXPERIENCE

Bloomberg LP | New York, New York — July 2022 – Present

Senior Software Engineer

- Architected a multi-language SDK for Bloomberg's centralized enterprise event bus, providing consistent publish/consume APIs across three programming languages to standardize messaging workflows across buy-side and sell-side platforms.
- Built event-driven trade capture pipelines processing millions of trades per day across buy-side and sell-side systems, integrating normalized trade events into downstream post-trade workflows and extending position-impact processing from sell-side-only to support buy-side activity.
- Architected and deployed high-throughput stream aggregation pipelines via Apache Kafka and Flink, processing 1M messages per second to power real-time market liquidity discovery and broker analytics.
- Engineered the corresponding Bloomberg analytics dashboard for historical trade indication (IOI) analysis, enabling traders to identify optimal brokers across securities and sectors using customizable metrics.
- Led a team of 5 to architect an MCP (Model Context Protocol) server, enabling LLMs to automate broker performance benchmarking by synthesizing trade indication aggregations with disparate financial data sources.
- Architected the organization's first Lakehouse and mentored an intern in building the end-to-end data pipeline, from Spark-based data movement and ingestion into Iceberg to analytics through Trino.
- Developed a real-time observability dashboard using Apache Superset and Flink to track LLM usage and cost attribution, enabling stakeholders to optimize AI resource allocation and mitigate budget overages.

Capital One | Richmond, Virginia — Jun 2021 – Aug 2021

Software Engineer Intern

- Engineered a serverless data pipeline using AWS Lambda to ingest 2M+ call routing profiles and queue status into DynamoDB; streamlined production deployment via internal CI/CD pipelines and staged rollouts.
- Automated SLA reporting for 12 global call centers by developing a dynamic management dashboard, enabling real-time visibility into queue performance and resource bottlenecks.

Gridology Education & Technology Inc. | New York, New York — May 2020 – Aug 2020

Software Engineer Intern

- Developed web scrapers to extract course information from university websites using a hybrid BeautifulSoup and Selenium approach.
- Reduced scraper runtime from 2 hours to 10 minutes by using BeautifulSoup for static pages and reserving Selenium only for sites using dynamic Javascript rendering.
- Updated Django REST APIs to match PostgreSQL schema changes, improving data integrity and query performance.

Hutyra Research Lab | Boston University — Feb 2020 – May 2020

Research Assistant

- Designed and developed an interactive web visualization platform for real-time monitoring and management of a distributed sensor network, enabling the study of carbon dynamics in forest systems.
- Created and deployed an HTTP server to efficiently ingest thousands of daily IoT sensor data points via the MQTT protocol, with long-term data persistence in MongoDB, enhancing data accessibility and scalability.

SKILLS

***Programming*:** Python, C++, C, TypeScript, JavaScript, Kotlin, Go, SQL, HTML/CSS, Bash

Databases: Iceberg, PostgreSQL, SQLite, MongoDB, DynamoDB, Redis

***Tools & Platforms*:** Kafka, Flink, Spark, Trino, Jenkins, Kubernetes, Docker, AWS, GCP, Firebase, OpenShift, Git

EDUCATION

Boston University | Boston, MA — May 2022

Bachelor of Science in Computer Engineering (**GPA**: 3.82/4.00; **Honors**: Dean's List, Albert Shanker Scholarship)